

Task 5: Exploratory Data Analysis (EDA) - Titanic Dataset

Objective

To explore the Titanic dataset using statistical analysis and visualization to identify patterns, relationships, trends, and anomalies.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme()
```

```
df = pd.read_csv("/train.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
```

```
11 Embarked      889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208	
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429	
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200	

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 891
Number of columns: 12
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

```
dtype: int64
```

```
print("Gender:")
print(df["Sex"].value_counts())

print("\nPassenger Class:")
print(df["Pclass"].value_counts())
```

```
print("\nSurvival:")
print(df["Survived"].value_counts())
```

```
Gender:
Sex
male    577
female  314
Name: count, dtype: int64
```

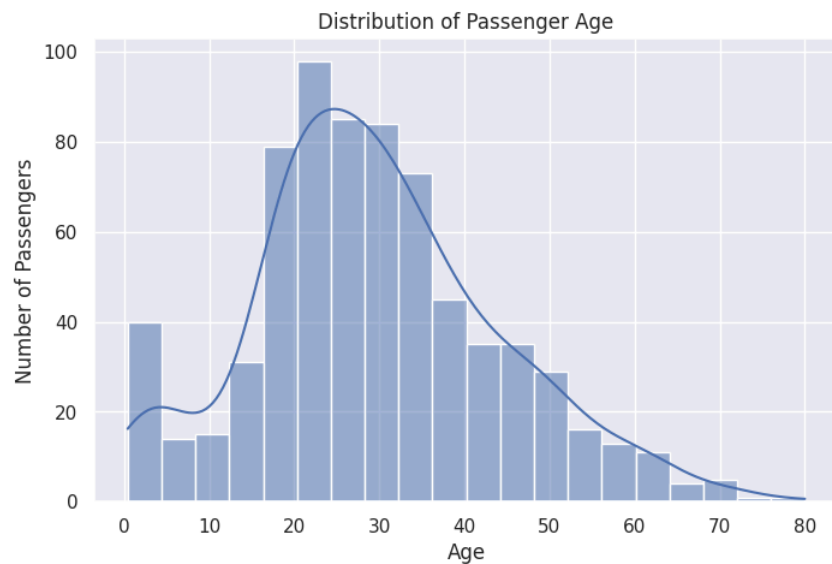
```
Passenger Class:
Pclass
3     491
1     216
2     184
Name: count, dtype: int64
```

```
Survival:
Survived
0     549
1     342
Name: count, dtype: int64
```

```
plt.figure(figsize=(8,5))

sns.histplot(df["Age"].dropna(), bins=20, kde=True)

plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



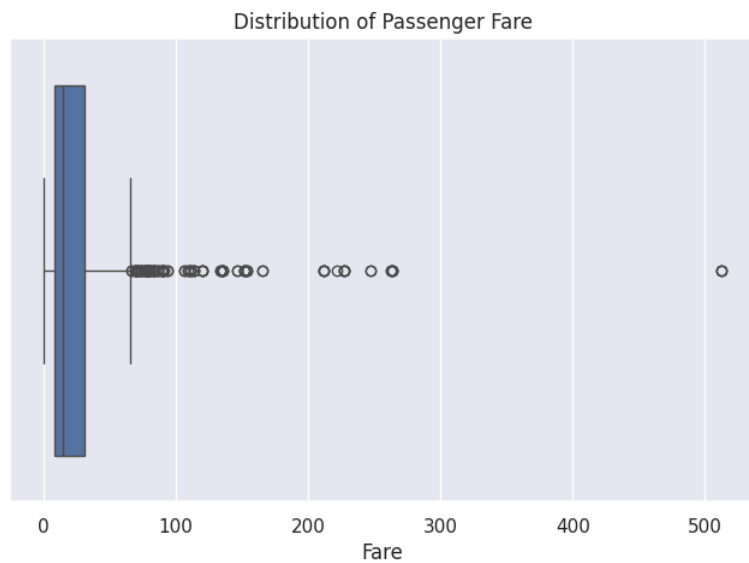
Observation

The histogram shows the distribution of passenger ages. Most passengers were young or middle-aged, while relatively fewer passengers were elderly.

```
plt.figure(figsize=(8,5))

sns.boxplot(x=df["Fare"])

plt.title("Distribution of Passenger Fare")
plt.xlabel("Fare")
plt.show()
```



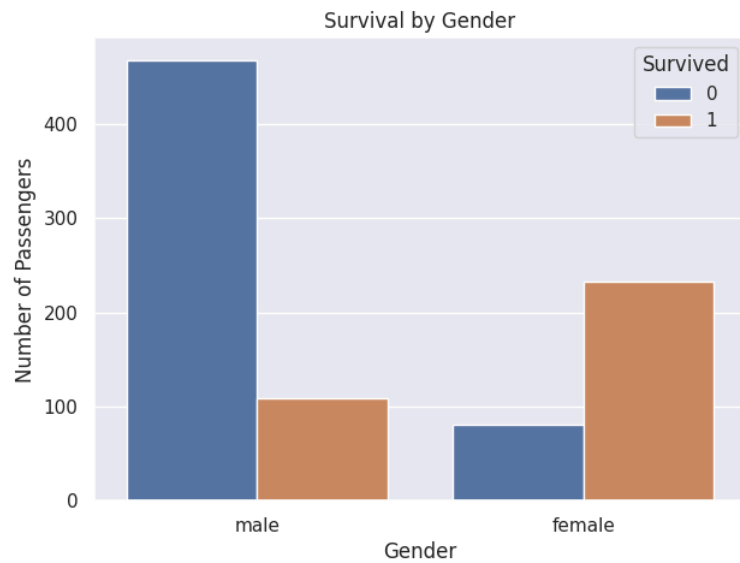
✓ Observation

The boxplot shows that most passengers paid relatively lower fares. A few passengers paid very high fares, indicating the presence of outliers.

```
plt.figure(figsize=(7,5))

sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



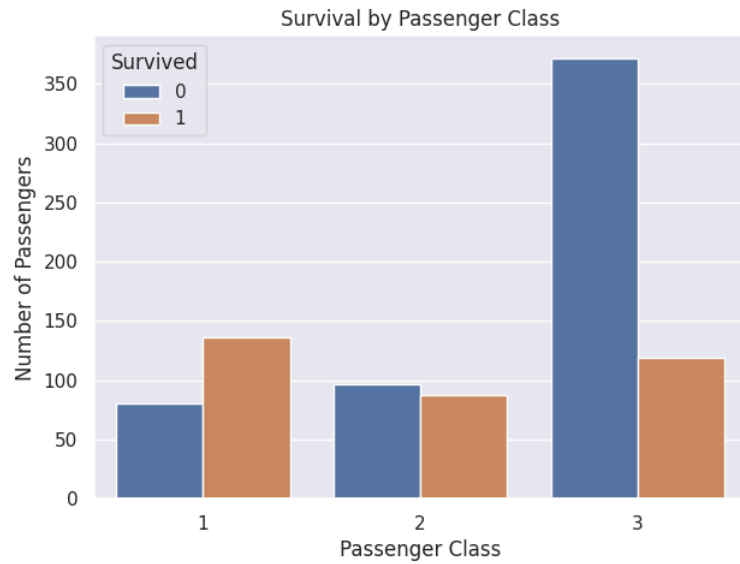
Observation

The graph shows a clear difference in survival between genders. Female passengers had a higher proportion of survivors compared with male passengers.

```
plt.figure(figsize=(7,5))

sns.countplot(data=df, x="Pclass", hue="Survived")

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



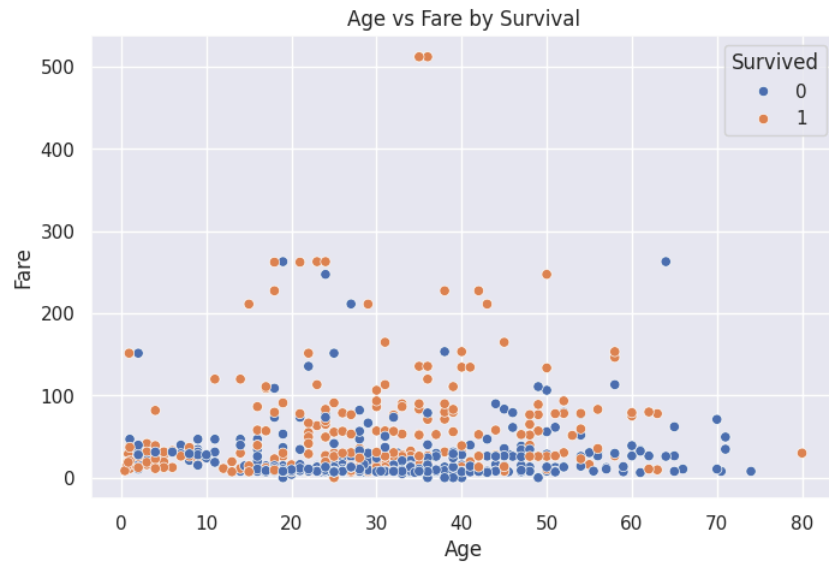
Observation

The graph shows that passengers in higher classes generally had better survival outcomes than passengers in lower classes.

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x="Age",
    y="Fare",
    hue="Survived"
)

plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



Observation

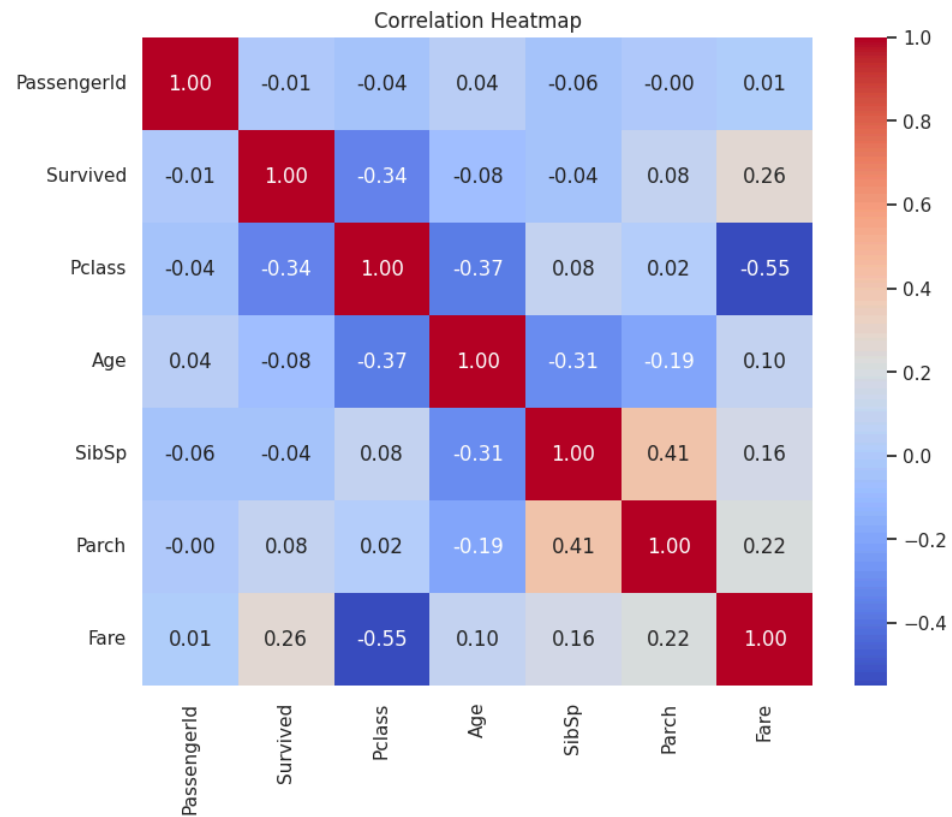
The scatterplot shows the relationship between passenger age and fare. There is considerable variation in fare values, while some passengers paid substantially higher fares than others.

```
plt.figure(figsize=(9,7))

numeric_df = df.select_dtypes(include=np.number)

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")
plt.show()
```



Observation

The heatmap shows the correlation between numerical variables. Passenger class and fare show a noticeable relationship, while survival also has relationships with some passenger characteristics.

```
pairplot_data = df[
    ["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]
].dropna()

sns.pairplot(
    pairplot_data,
    hue="Survived"
)

plt.show()
```

